

SET 3 — ANDROID STUDIO

Q11–Q15 COMPLETE COPY-PASTE GUIDE

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Use the exact file names and paste the complete code shown for each file.

SET 3 — ANDROID STUDIO

Q11–Q15 COMPLETE COPY-PASTE CODE GUIDE

Purpose: This guide contains only the complete code, exact file locations, and exactly where to paste each part. It is designed for quick practical-exam use.

Important: For every file below, open the named file and replace its existing contents with the complete code shown. Do not paste one block inside another block.

Q11 — Temperature Converter — LinearLayout

1. Open: app → res → layout → activity_main.xml Replace EVERYTHING with the XML below.
2. Open: app → res → values → strings.xml Replace EVERYTHING with the strings below.
3. Open: app → res → values → colors.xml Replace EVERYTHING with the colors below.
4. Open: app → res → values → dims.xml Replace EVERYTHING with the dimensions below.
5. Open: app → java/kotlin → YOUR PACKAGE → MainActivity.kt Keep your own package line if it is different, then replace the rest with the code below.
6. Press Run ■ and test both conversion options.

FILE: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="@dimen/screen_padding"
    android:background="@color/screen_background">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/title_temperature"
        android:textSize="@dimen/title_size"
        android:textColor="@color/title_color"
        android:gravity="center"
        android:layout_marginBottom="@dimen/large_margin" />

    <EditText
        android:id="@+id/etTemperature"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/temperature_hint"
        android:inputType="numberDecimal|numberSigned"
        android:textSize="@dimen/text_size" />

    <RadioGroup
```

```
    android:id="@+id/radioGroup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginTop="@dimen/medium_margin">
```

```
<RadioButton
```

```
    android:id="@+id/rbCtoF"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/c_to_f"
    android:textSize="@dimen/text_size" />
```

```
<RadioButton
```

```
    android:id="@+id/rbFtoC"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/f_to_c"
    android:textSize="@dimen/text_size" />
```

```
</RadioGroup>
```

```
<Button
```

```
    android:id="@+id/btnConvert"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/convert"
    android:layout_marginTop="@dimen/medium_margin" />
```

```
<TextView
```

```
    android:id="@+id/tvResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/result_default"
    android:textSize="@dimen/result_size"
    android:gravity="center"
    android:layout_marginTop="@dimen/large_margin" />
```

```
</LinearLayout>
```

FILE: strings.xml

```
<resources>
```

```
    <string name="app_name">Temperature Converter</string>
```

```
    <string name="title_temperature">Temperature Converter</string>
```

```

<string name="temperature_hint">Enter temperature</string>
<string name="c_to_f">Celsius to Fahrenheit</string>
<string name="f_to_c">Fahrenheit to Celsius</string>
<string name="convert">Convert</string>
<string name="result_default">Result will appear here</string>
<string name="empty_error">Please enter a temperature.</string>
<string name="selection_error">Please select a conversion type.</string>

```

</resources>

FILE: colors.xml

```

<resources>
    <color name="screen_background">#FFFFFF</color>
    <color name="title_color">#222222</color>
    <color name="default_result">#222222</color>
    <color name="c_to_f_result">#1565C0</color>
    <color name="f_to_c_result">#2E7D32</color>

```

</resources>

FILE: dimens.xml

```

<resources>
    <dimen name="screen_padding">16dp</dimen>
    <dimen name="title_size">24sp</dimen>
    <dimen name="text_size">18sp</dimen>
    <dimen name="result_size">22sp</dimen>
    <dimen name="medium_margin">12dp</dimen>
    <dimen name="large_margin">24dp</dimen>

```

</resources>

FILE: MainActivity.kt

```

package YOUR.PACKAGE.NAME

import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.RadioButton
import android.widget.RadioGroup
import android.widget.TextView
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity
import androidx.core.content.ContextCompat
import java.util.Locale

class MainActivity : AppCompatActivity() {

```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)
    val etTemperature = findViewById<EditText>(R.id.etTemperature)
    val radioGroup = findViewById<RadioGroup>(R.id.radioGroup)
    val rbCtoF = findViewById<RadioButton>(R.id.rbCtoF)
    val rbFtoC = findViewById<RadioButton>(R.id.rbFtoC)
    val btnConvert = findViewById<Button>(R.id.btnConvert)

    val tvResult = findViewById<TextView>(R.id.tvResult)
    btnConvert.setOnClickListener {
        val input = etTemperature.text.toString().trim()
        if (input.isEmpty()) {
            AlertDialog.Builder(this)
                .setTitle("Error")
                .setMessage(getString(R.string.empty_error))
                .setPositiveButton("OK", null)
                .show()
            return@setOnClickListener
        }
        if (radioGroup.checkedRadioButtonId == -1) {
            AlertDialog.Builder(this)
                .setTitle("Error")
                .setMessage(getString(R.string.selection_error))
                .setPositiveButton("OK", null)
                .show()
            return@setOnClickListener
        }
        val temperature = input.toDoubleOrNull()
        if (temperature == null) {
            AlertDialog.Builder(this)
                .setTitle("Error")
                .setMessage("Please enter a valid number.")
                .setPositiveButton("OK", null)
                .show()
            return@setOnClickListener
        }
        if (rbCtoF.isChecked) {

```

```

        val result = (temperature * 9 / 5) + 32
        tvResult.text = String.format(
            Locale.getDefault(), "%.2f °F", result
        )
        tvResult.setTextColor(
            ContextCompat.getColor(this, R.color.c_to_f_result)
        )
    } else if (rbFtoC.isChecked) {
        val result = (temperature - 32) * 5 / 9
        tvResult.text = String.format(
            Locale.getDefault(), "%.2f °C", result
        )
        tvResult.setTextColor(
            ContextCompat.getColor(this, R.color.f_to_c_result)
        )
    }
}
}
}
}

```

FINAL RUN CHECK

Press **Build → Make Project**. If there are no errors, press **Run ■**.

Select your

emulator/phone. Test the required valid case and the required error case.

Q12 — Feedback Screen — AbsoluteLayout

1. Open: app → res → layout → activity_main.xml Replace EVERYTHING. 2. Open: app → res → values → strings.xml Replace EVERYTHING. 3. Open: app → res → values → colors.xml Replace EVERYTHING. 4. Open: app → res → values → dimens.xml Replace EVERYTHING. 5. Open: MainActivity.kt Replace EVERYTHING except change YOUR.PACKAGE.NAME to your actual package. 6. Run the app. Enter a name and comments, then press Submit.

FILE: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/screen_background">

    <TextView

        android:id="@+id/tvHeading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/feedback_title"
        android:textSize="@dimen/title_size"
        android:textColor="@color/title_color"
        android:layout_x="100dp"
        android:layout_y="30dp" />

    <EditText

        android:id="@+id/etName"
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:hint="@string/name_hint"
        android:textSize="@dimen/text_size"
        android:layout_x="40dp"
        android:layout_y="100dp" />

    <EditText

        android:id="@+id/etComments"
        android:layout_width="300dp"
        android:layout_height="120dp"
```

```
    android:hint="@string/comments_hint"
    android:gravity="top"
    android:inputType="textMultiLine"
    android:textSize="@dimen/text_size"
    android:layout_x="40dp"
    android:layout_y="170dp" />
```

<Button

```
    android:id="@+id/btnSubmit"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:text="@string/submit"
    android:layout_x="115dp"
    android:layout_y="320dp" />
```

<TextView

```
    android:id="@+id/tvOutput"
    android:layout_width="340dp"
    android:layout_height="wrap_content"
    android:textSize="@dimen/text_size"
    android:textColor="@color/output_color"
    android:layout_x="30dp"
    android:layout_y="400dp" />
```

</AbsoluteLayout>

FILE: strings.xml

<resources>

```
<string name="app_name">Feedback Screen</string>
<string name="feedback_title">Feedback</string>
<string name="name_hint">Enter your name</string>
<string name="comments_hint">Enter your comments</string>
<string name="submit">Submit</string>
```

</resources>

FILE: colors.xml

<resources>

```
<color name="screen_background">#FFFFFF</color>
<color name="title_color">#222222</color>
<color name="output_color">#222222</color>
```

</resources>

FILE: dimens.xml

```
<resources>
```

```
    <dimen name="title_size">24sp</dimen>
```

```
    <dimen name="text_size">18sp</dimen>
```

```
</resources>
```

```
FILE: MainActivity.kt
```

```
package YOUR.PACKAGE.NAME
```

```
import android.os.Bundle
```

```
import android.widget.Button
```

```
import android.widget.EditText
```

```
import android.widget.TextView
```

```
import android.widget.Toast
```

```
import androidx.appcompat.app.AlertDialog
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        val etName = findViewById<EditText>(R.id.etName)
```

```
        val etComments = findViewById<EditText>(R.id.etComments)
```

```
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)
```

```
        val tvOutput = findViewById<TextView>(R.id.tvOutput)
```

```
        btnSubmit.setOnClickListener {
```

```
            val name = etName.text.toString().trim()
```

```
            val comments = etComments.text.toString().trim()
```

```
            if (name.isEmpty()) {
```

```
                AlertDialog.Builder(this)
```

```
                    .setTitle("Error")
```

```
                    .setMessage("Please enter your name.")
```

```
                    .setPositiveButton("OK", null)
```

```
                    .show()
```

```
                return@setOnClickListener
```

```
            }
```

```
            tvOutput.text = "Name: $name\nComments: $comments"
```

```
            Toast.makeText(
```

```
                this,
```

```
                "Thank you",
```

```
                Toast.LENGTH_SHORT
```

```
            ).show()
```

```
    }  
  }  
}
```

FINAL RUN CHECK

Press **Build → Make Project**. If there are no errors, press **Run ■**.
Select your
emulator/phone. Test the required valid case and the required error case.

Q13 — Rating Screen — RelativeLayout

1. Open: activity_main.xml → replace EVERYTHING.
2. Open: strings.xml → replace EVERYTHING.
3. Open: colors.xml → replace EVERYTHING.
4. Open: dims.xml → replace EVERYTHING.
5. Open: MainActivity.kt → replace EVERYTHING except your package name.
6. Run. Give a rating and comments, then press Submit.

FILE: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="@dimen/screen_padding"
    android:background="@color/screen_background">

    <TextView

        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/rating_title"
        android:textSize="@dimen/title_size"
        android:textColor="@color/title_color"
        android:layout_centerHorizontal="true" />

    <RatingBar

        android:id="@+id/ratingBar"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:numStars="5"
        android:stepSize="1"
        android:layout_below="@id/tvTitle"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="@dimen/large_margin" />

    <EditText

        android:id="@+id/etComments"
        android:layout_width="match_parent"
        android:layout_height="120dp"
        android:hint="@string/comments_hint"
        android:gravity="top"
```

```

        android:inputType="textMultiLine"
        android:layout_below="@id/ratingBar"
        android:layout_marginTop="@dimen/medium_margin" />
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/submit"
    android:layout_below="@id/etComments"
    android:layout_marginTop="@dimen/medium_margin" />
<TextView
    android:id="@+id/tvOutput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="@dimen/text_size"
    android:layout_below="@id/btnSubmit"
    android:layout_marginTop="@dimen/large_margin" />
</RelativeLayout>

```

FILE: strings.xml

```

<resources>
    <string name="app_name">Rating Screen</string>
    <string name="rating_title">Rate Your Experience</string>

    <string name="comments_hint">Enter your comments</string>
    <string name="submit">Submit</string>
</resources>

```

FILE: colors.xml

```

<resources>
    <color name="screen_background">#FFFFFF</color>
    <color name="title_color">#222222</color>
    <color name="good_color">#2E7D32</color>
    <color name="bad_color">#D32F2F</color>
</resources>

```

FILE: dimens.xml

```

<resources>
    <dimen name="screen_padding">16dp</dimen>
    <dimen name="title_size">24sp</dimen>
    <dimen name="text_size">18sp</dimen>

```

```
<dimen name="medium_margin">12dp</dimen>
```

```
<dimen name="large_margin">24dp</dimen>
```

```
</resources>
```

FILE: MainActivity.kt

```
package YOUR.PACKAGE.NAME
```

```
import android.os.Bundle
```

```
import android.widget.Button
```

```
import android.widget.EditText
```

```
import android.widget.RatingBar
```

```
import android.widget.TextView
```

```
import androidx.appcompat.app.AlertDialog
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
import androidx.core.content.ContextCompat
```

```
class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        val ratingBar = findViewById<RatingBar>(R.id.ratingBar)
```

```
        val etComments = findViewById<EditText>(R.id.etComments)
```

```
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)
```

```
        val tvOutput = findViewById<TextView>(R.id.tvOutput)
```

```
        btnSubmit.setOnClickListener {
```

```
            val rating = ratingBar.rating
```

```
            val comments = etComments.text.toString().trim()
```

```
            tvOutput.text = "Rating: ${rating.toInt()} / 5\nComments: $comments"
```

```
            if (rating < 3) {
```

```
                tvOutput.setTextColor(
```

```
                    ContextCompat.getColor(this, R.color.bad_color)
```

```
                )
```

```
                AlertDialog.Builder(this)
```

```
                    .setTitle("Low Rating")
```

```
                    .setMessage("Would you like to add more details?")
```

```
                    .setPositiveButton("Yes", null)
```

```
                    .setNegativeButton("No", null)
```

```
                    .show()
```

```
            } else {
```

```
                tvOutput.setTextColor(
```

```
                    ContextCompat.getColor(this, R.color.good_color)
```

```
    )  
    android.widget.Toast.makeText(  
        this,  
        "Thank you for your rating!",  
        android.widget.Toast.LENGTH_SHORT  
    ).show()  
}  
}  
}
```

FINAL RUN CHECK

Press **Build → Make Project**. If there are no errors, press **Run** . Select your emulator/phone. Test the required valid case and the required error case.

Q14 — Profile Card — LinearLayout

1. Open: activity_main.xml → replace EVERYTHING. 2. Open: strings.xml → replace EVERYTHING. 3. Open: colors.xml → replace EVERYTHING. 4. Open: dimens.xml → replace

EVERYTHING. 5. Put an image named profile_image.png/jpg in: app → res → drawable 6.

Open: MainActivity.kt → replace EVERYTHING except your package name. 7. Run. Test

Switch Mode and Contact. 8. If your project already has a generated themes.xml, keep the

generated theme. This practical's mode switching is handled directly in Kotlin.

FILE: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/mainLayout"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="@dimen/screen_padding"

    android:background="@color/light_background">

    <ImageView

        android:id="@+id/profileImage"

        android:layout_width="@dimen/image_size"

        android:layout_height="@dimen/image_size"

        android:src="@drawable/profile_image"

        android:scaleType="centerCrop"

        android:contentDescription="@string/profile_image_desc" />

    <TextView

        android:id="@+id/tvName"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="@string/profile_name"

        android:textSize="@dimen/title_size"

        android:textColor="@color/light_text"

        android:textStyle="bold"

        android:layout_marginTop="@dimen/medium_margin" />

    <TextView
```

```
    android:id="@+id/tvCourse"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/profile_course"
    android:textSize="@dimen/text_size"
    android:textColor="@color/light_text"
    android:layout_marginTop="@dimen/small_margin" />
```

<TextView

```
    android:id="@+id/tvDescription"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/profile_description"
    android:textSize="@dimen/text_size"
    android:textColor="@color/light_text"
    android:gravity="center"
    android:layout_marginTop="@dimen/medium_margin" />
```

<Button

```
    android:id="@+id/btnMode"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/switch_mode"
    android:layout_marginTop="@dimen/medium_margin" />
```

<Button

```
    android:id="@+id/btnContact"

    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/contact"
    android:layout_marginTop="@dimen/small_margin" />
```

</LinearLayout>

FILE: strings.xml

<resources>

```
    <string name="app_name">Profile Card</string>
    <string name="profile_image_desc">Profile Image</string>
    <string name="profile_name">Your Name</string>
    <string name="profile_course">BSc IT</string>
    <string name="profile_description">Android Programming Student</string>
    <string name="switch_mode">Switch Mode</string>
```

```
    <string name="contact">Contact</string>
</resources>
```

FILE: colors.xml

```
<resources>

    <color name="light_background">#FFFFFF</color>

    <color name="dark_background">#212121</color>

    <color name="light_text">#222222</color>

    <color name="dark_text">#FFFFFF</color>

</resources>
```

FILE: dims.xml

```
<resources>

    <dimen name="screen_padding">20dp</dimen>

    <dimen name="image_size">140dp</dimen>

    <dimen name="title_size">24sp</dimen>

    <dimen name="text_size">18sp</dimen>

    <dimen name="small_margin">8dp</dimen>

    <dimen name="medium_margin">16dp</dimen>

</resources>
```

FILE: MainActivity.kt

```
package YOUR.PACKAGE.NAME

import android.os.Bundle
import android.widget.Button
import android.widget.LinearLayout
import android.widget.TextView
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity
import androidx.core.content.ContextCompat

class MainActivity : AppCompatActivity() {

    private var darkMode = false

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val mainLayout = findViewById<LinearLayout>(R.id.mainLayout)

        val tvName = findViewById<TextView>(R.id.tvName)

        val tvCourse = findViewById<TextView>(R.id.tvCourse)

        val tvDescription = findViewById<TextView>(R.id.tvDescription)

        val btnMode = findViewById<Button>(R.id.btnMode)

        val btnContact = findViewById<Button>(R.id.btnContact)
```

```

btnMode.setOnClickListener {
    darkMode = !darkMode
    if (darkMode) {

        mainLayout.setBackgroundColor(
            ContextCompat.getColor(this, R.color.dark_background)
        )
        tvName.setTextColor(
            ContextCompat.getColor(this, R.color.dark_text)
        )
        tvCourse.setTextColor(
            ContextCompat.getColor(this, R.color.dark_text)
        )
        tvDescription.setTextColor(
            ContextCompat.getColor(this, R.color.dark_text)
        )
    } else {
        mainLayout.setBackgroundColor(
            ContextCompat.getColor(this, R.color.light_background)
        )
        tvName.setTextColor(
            ContextCompat.getColor(this, R.color.light_text)
        )
        tvCourse.setTextColor(
            ContextCompat.getColor(this, R.color.light_text)
        )
        tvDescription.setTextColor(
            ContextCompat.getColor(this, R.color.light_text)
        )
    }
}

btnContact.setOnClickListener {
    AlertDialog.Builder(this)
        .setTitle("Contact")
        .setMessage("Email: example@gmail.com\nPhone: 9876543210")
        .setPositiveButton("OK", null)
        .show()
}

```

```
}
```

```
}
```

Q14 image: Put your image file inside **app → res → drawable**. The filename must match **profile_image** (for example, **profile_image.png**). If your image has another filename, change the **@drawable/profile_image** line in **activity_main.xml** accordingly.

FINAL RUN CHECK

Press **Build → Make Project**. If there are no errors, press **Run** . Select your emulator/phone. Test the required valid case and the required error case.

Q15 — Multiplication Table — TableLayout

1. Open: activity_main.xml → replace EVERYTHING. 2. Open: strings.xml → replace EVERYTHING. 3. Open: colors.xml → replace EVERYTHING. 4. Open: dims.xml → replace EVERYTHING. 5. Open: MainActivity.kt → replace EVERYTHING except your package name. 6. Run. Enter a non-zero number and press Generate. 7. Press Clear to remove the generated table.

FILE: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:padding="@dimen/screen_padding">

    <EditText

        android:id="@+id/etNumber"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="@string/number_hint"

        android:inputType="numberSigned"

        android:textSize="@dimen/text_size" />

    <Button

        android:id="@+id/btnGenerate"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="@string/generate"

        android:layout_marginTop="@dimen/medium_margin" />

    <TableLayout

        android:id="@+id/tableLayout"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:stretchColumns="0"

        android:layout_marginTop="@dimen/medium_margin" />

    <Button

        android:id="@+id/btnClear"

        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"

        android:text="@string/clear"

        android:layout_marginTop="@dimen/medium_margin" />
</LinearLayout>

```

FILE: strings.xml

```

<resources>

    <string name="app_name">Multiplication Table</string>

    <string name="number_hint">Enter number</string>

    <string name="generate">Generate</string>

    <string name="clear">Clear</string>

</resources>

```

FILE: colors.xml

```

<resources>

    <color name="row_color_1">#E3F2FD</color>

    <color name="row_color_2">#FFFFFF</color>

</resources>

```

FILE: dimens.xml

```

<resources>

    <dimen name="screen_padding">16dp</dimen>

    <dimen name="text_size">18sp</dimen>

    <dimen name="medium_margin">12dp</dimen>

    <dimen name="row_padding">10dp</dimen>

</resources>

```

FILE: MainActivity.kt

```

package YOUR.PACKAGE.NAME

import android.graphics.Color
import android.os.Bundle
import android.view.Gravity
import android.widget.Button
import android.widget.EditText
import android.widget.TableLayout
import android.widget.TableRow
import android.widget.TextView
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity
import androidx.core.content.ContextCompat

class MainActivity : AppCompatActivity() {

```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)
    val etNumber = findViewById<EditText>(R.id.etNumber)
    val btnGenerate = findViewById<Button>(R.id.btnGenerate)
    val btnClear = findViewById<Button>(R.id.btnClear)
    val tableLayout = findViewById<TableLayout>(R.id.tableLayout)
    btnGenerate.setOnClickListener {
        val input = etNumber.text.toString().trim()
        if (input.isEmpty()) {
            AlertDialog.Builder(this)
                .setTitle("Error")
                .setMessage("Please enter a number.")
                .setPositiveButton("OK", null)
                .show()
            return@setOnClickListener
        }
        val number = input.toIntOrNull()
        if (number == null || number == 0) {
            AlertDialog.Builder(this)
                .setTitle("Error")
                .setMessage("Please enter a non-zero number.")
                .setPositiveButton("OK", null)
                .show()
            return@setOnClickListener
        }
        tableLayout.removeAllViews()
        for (i in 1..10) {
            val row = TableRow(this)
            val textView = TextView(this)
            textView.text = "$number x $i = ${number * i}"
            textView.textSize = 18f
            textView.setPadding(10, 10, 10, 10)
            textView.gravity = Gravity.CENTER
            if (i % 2 == 0) {
                row.setBackgroundColor(
                    ContextCompat.getColor(this, R.color.row_color_1)
                )
            }
            tableLayout.addView(row)
        }
    }
}

```

```

        )
    } else {
        row.setBackgroundColor(
            ContextCompat.getColor(this, R.color.row_color_2)
        )
    }
    row.addView(textView)
    tableLayout.addView(row)
}
}
btnClear.setOnClickListener {
    etNumber.text.clear()
    tableLayout.removeAllViews()
}
}
}

```

FINAL RUN CHECK

Press **Build → Make Project**. If there are no errors, press **Run ■**.
 Select your
 emulator/phone. Test the required valid case and the required error case.

EXAM QUICK RULES

1. Always replace the complete contents of the named file. Do not paste XML into

Kotlin or Kotlin into XML.

2. In MainActivity.kt, replace **YOUR.PACKAGE.NAME** with the package name

already shown at the top of your existing MainActivity.kt.

3. If Android Studio shows **Unresolved reference: R**, first check the XML files

for red errors and run Build → Make Project.

4. Q14 requires an image in the drawable folder.

5. Q15 automatically creates rows 1–10 when Generate is pressed.

6. The code is organized so each practical can be implemented as a separate Android

Studio project or by replacing the files in the current project.

SET 3 COMPLETE — Q11, Q12, Q13, Q14,

Q15